Report

Of

Data Analysis

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410030100

Name of Student: DIBYESH BISWAL

Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I **Dibyesh Biswal** do hereby declare that the internship report titled **"DATA ANALYSIS ESSENTIALS"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature
Student Name- Dibyesh Biswal
Roll No.2410030100

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.
I express my gratitude to my parents for their blessings.

Signature
Student Name – Dibyesh Biswal
Roll No..2410030100

Cisco
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This certificate is awarded to

Dibyesh Biswal

for successfully completing

Data Analytics Essentials

offered by IILM University
through the Cisco Networking Academy program.

Swati Vashisht

Swati Vashisht
Instructor
IILM University

16 Jun 2026
Completion Date

4.PROJECT DISCRIPTION

4.1 INTRODUCTION

Data Analytics is an important field that helps organizations understand data and make better decisions. During my internship, I gained practical knowledge about the basic concepts and techniques used in Data Analytics. The internship provided me with an opportunity to understand how raw data can be collected, processed, analyzed, and presented in a meaningful way.

The main focus of the internship was to develop an understanding of data analysis concepts and tools. I learned about data handling, data exploration, visualization, and the process of extracting useful information from datasets. This practical exposure helped me connect the theoretical concepts learned in academics with real-world applications.

Overall, the internship improved my technical knowledge, analytical thinking, and problem-solving skills. It also provided valuable experience in working with data and helped me understand the importance of Data Analytics in different fields.

4.2 Organization Profile

The organization provides training and practical learning opportunities in the field of Data Analytics and technology. It helps students develop industry-relevant skills through structured training, hands-on activities, and practical applications. The internship provided valuable exposure to data handling, analysis, and visualization.

4.3 Problem Statement

Organizations generate large amounts of data, but extracting useful information from this data can be challenging. The problem is to analyze, organize, and visualize data effectively to identify meaningful patterns and support better decision-making.

4.4 Project Objectives

The main objective of this internship is to understand the process of Data Analytics and learn how raw data can be converted into useful information. The project aims to analyze and organize the available data, identify important patterns and trends, and present the findings through effective data visualization. It also focuses on developing practical skills in data handling, analysis, and interpretation. Through this internship, the objective is to improve analytical and problem-solving skills and understand how data-driven insights can support better decision-making.

4.5 Scope of the Project

The scope of this project is to develop a clear understanding of the Data Analytics process and its practical applications. The project focuses on working with data from the initial stage of collection and organization to analysis and visualization. It includes understanding how raw and unstructured data can be cleaned, processed, and transformed into meaningful information. Data analysis techniques are used to identify important patterns, trends, relationships, and useful insights from the available dataset.

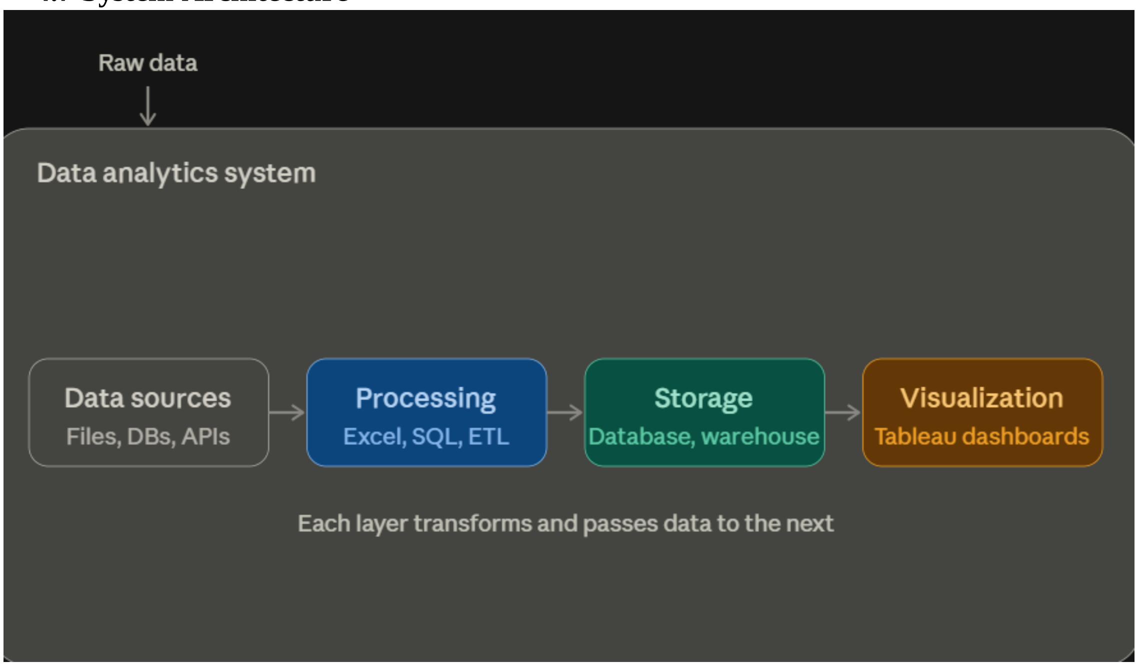
The project also covers the use of data visualization techniques to represent analytical results in a simple and understandable manner. Different charts, graphs, and visual representations can be used to make complex data easier to interpret. Through this project, practical knowledge of data handling, exploration, analysis, and interpretation can be developed.

Furthermore, the project provides an opportunity to improve technical, analytical, and problem-solving skills by working with real-world data. The knowledge gained during the internship can be useful for making data-driven decisions and can also provide a foundation for further learning in areas such as advanced analytics, machine learning, and business intelligence.

4.6 Technologies and Tools Used

The project involved the use of various technologies and tools related to Data Analytics. **Python** was used for data processing, analysis, and handling datasets. **Pandas and NumPy** were used for data manipulation and numerical operations. **Matplotlib and Seaborn** were used to create charts and graphs for data visualization. **Jupyter Notebook** was used as the working environment for writing and executing Python code. These tools helped in cleaning, analyzing, exploring, and presenting the data in an organized and understandable form.

4.7 System Architecture



4.8 METHODOLOGY

The methodology followed in this project consists of a systematic process for analyzing and interpreting data. First, the required dataset is collected and examined to understand its structure and important features. The collected data is then cleaned by identifying and handling missing values, duplicate records, and incorrect or inconsistent information. After data cleaning, the dataset is organized and prepared for further analysis.

The next step involves **data exploration**, where different variables are studied to identify important patterns, trends, and relationships within the data. Appropriate statistical and analytical techniques are applied to obtain meaningful insights. Data visualization is then performed using suitable charts and graphs to represent the findings in a simple and understandable manner.

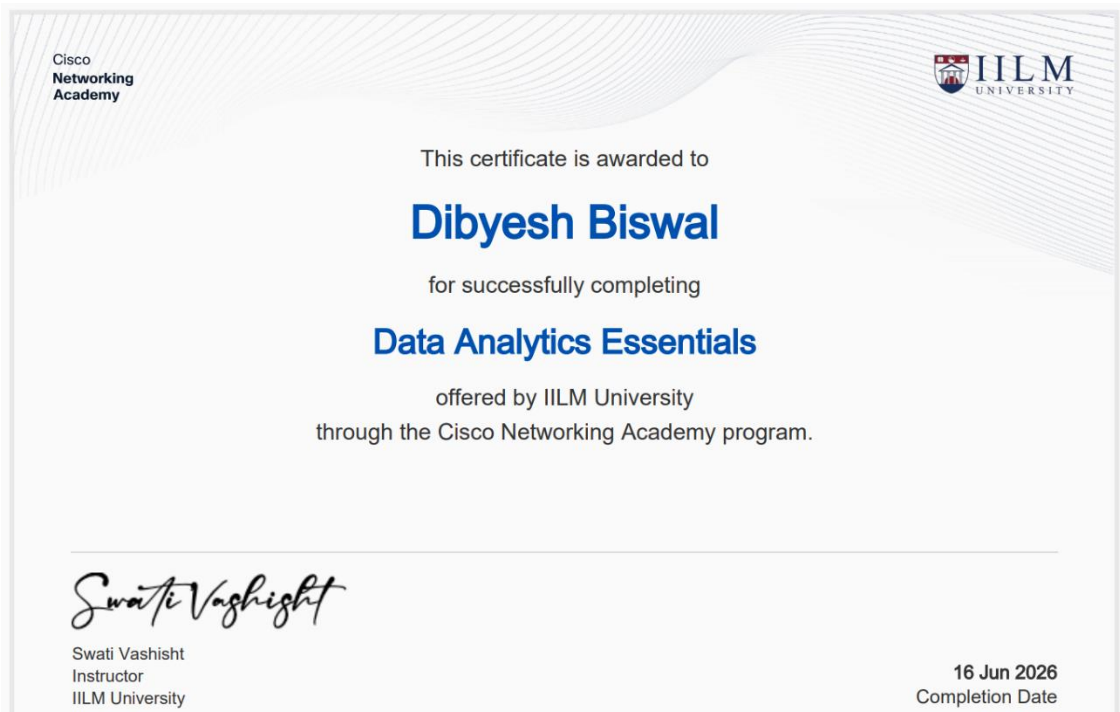
Finally, the analyzed results are interpreted to identify useful insights and support data-driven decision-making. This methodology helps in converting raw data into meaningful information while developing practical skills in data preparation, analysis, visualization, and interpretation.

4.9 EXPECTED OUTCOMES

The expected outcome of this project is to develop a clear understanding of the Data Analytics process and its practical applications. The project is expected to provide practical knowledge of data collection, cleaning, processing, exploration, analysis, and visualization. It will help in identifying meaningful patterns and trends from the available data and presenting the results in an understandable form.

Through this internship, the project is also expected to improve technical and analytical skills, including data handling, problem-solving, and interpretation of results. The use of different analytical and visualization techniques will help in converting raw data into useful information. Overall, the project is expected to provide practical exposure to Data Analytics and build a strong foundation for future work in data-driven applications and decision-making.

4.10 Certificates of completion and other communication mail proof



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Swati Vashisht
Instructor
IILM University

16 Jun 2026
Completion Date

Networking Academy

Certificate of Course Completion

Dibyesh Biswal

*has successfully achieved student level credential for completing the **Data Analytics Essentials** course.*

The student was able to proficiently:

- *Explain how the data analytics process creates value from data.*
- *Explain the characteristics of data, including formats, availability and methods to acquire.*
- *Transform data using analytics tools.*
- *Analyze data using basic statistical and data preparation techniques.*
- *Complete hands-on lab using Excel, SQL, Tableau and other tools.*
- *Evaluate and share project portfolio.*

Scan to Verify

Lynn Bloomer

Lynn Bloomer
Director, Cisco Networking Academy

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